Notes: Qian, Strahan, and Yang (JF, 2015)

The Impact of Incentives and Communication Costs on Information Production and Use: Evidence from Bank Lending

Contents

1 Big picture

- **Question.** How do incentives and communication frictions inside a bank affect the quality of information that gets produced and the way that information is used in lending decisions?
- **Setting.** The paper studies one large nationwide Chinese state-owned bank using proprietary loan-level data from 2000 to 2006, matched to information on individual loan officers and branch presidents.
- **Core idea.** Better internal organization should improve both *information production* and *information use*. If loan officers are personally accountable for ratings, and if they can communicate more easily with the branch president who approves loans, then the bank should rely more on internally generated ratings and those ratings should better predict ex post repayment.
- **Two mechanisms.**
  - *Incentives*: a reform in 2002–2003 delegated responsibility from committees to identifiable individuals, especially the loan officers producing credit ratings.
  - *Communication costs*: after reform, some loan officer–branch president pairs had worked together longer than others. Longer overlap is used as a proxy for lower communication costs.
- **Main empirical test.** Does the bank place more weight on its internal credit rating in pricing loans and in forecasting default after the reform, and when the loan officer and branch president have worked together longer?
- **Main baseline result.** Yes. After reform, a safer internal rating lowers interest rates by more than before, and the rating becomes a stronger predictor of eventual repayment.
- **Main communication-cost result.** In the post-reform period, the effect of the internal rating on both loan pricing and loan outcomes becomes stronger when the loan officer and branch president have greater time overlap in the same branch.
- **Why this matters.** The paper is not only saying "ratings matter." It is saying that bank organization determines whether ratings are informative enough to matter. That connects theories of authority, incentives, and communication directly to loan contract terms and default outcomes.
- **Key interpretation.** Better internal incentives and lower communication costs improve the quality of soft-information production, allow that information to flow into pricing, and ultimately improve lending outcomes.
- **Takeaway.** Internal bank organization is part of credit allocation technology. The same observable borrower can be priced and monitored differently depending on who produces information, who approves the loan, and how well those people are connected inside the bank.

2 Data and measurement (key objects)

2.1 Institutional environment and reform

China's large state-owned banks historically relied on centralized and committee-based lending decisions. Following China's entry into the WTO in December 2001, many banks introduced reforms during the second half of 2002 and throughout 2003 that decentralized the lending process. Responsibility at each step of the lending process became more clearly assigned to individuals, and loan officers in the investigation unit became personally responsible for internal credit ratings.

This institutional change is crucial for the paper because it creates a plausibly exogenous increase in the incentives of loan officers to produce accurate information. The reforms did not arise from local loan-by-loan conditions; they were pushed from the top of the organization in response to external competitive pressure.

2.2 Main loan sample

The proprietary sample covers loans originated between January 2000 and December 2006. The authors define:

- **pre-reform period**: January 14, 2000 to April 16, 2002,
- **transition period dropped**: April 17, 2002 to December 31, 2003,
- **post-reform period**: January 1, 2004 to December 31, 2006.

The full sample contains 37,661 loans: 3,665 pre-reform and 33,996 post-reform. Borrowers come from 33 Chinese cities, and the bank operates through 330 branches pre-reform and 438 branches post-reform.

2.3 Borrowers and branches

The firm-level controls include:

- log assets,
- leverage,
- return on assets (ROA),
- ownership dummies, especially SOE and private enterprise,
- previous default record,
- total asset turnover,
- branch size, measured by total deposits in the initial year 2000.

Some key summary facts from Table I:

- mean firm assets rise from RMB 201 million pre-reform to RMB 354 million post-reform,
- mean leverage falls from 0.52 to 0.45,
- mean ROA rises from 0.06 to 0.09,

- mean loan size rises from RMB 4.13 million to RMB 6.63 million.

Interpretation. The post-reform sample occurs in a stronger macro period with larger firms and larger loans, so the empirical strategy must control carefully for borrower characteristics and time effects rather than comparing unconditional averages.

2.4 Internal credit rating

The core information measure is the bank's internal credit rating. It ranges from 1 to 8, where:

- 1 = riskiest borrower,
- 8 = safest borrower.

The rating combines hard information and soft information. Loan officers collect financial data, but they also use discussions with managers, guarantors, business partners, customers, and sometimes local officials. This makes the rating exactly the kind of information object whose quality should depend on both incentives and communication.

2.5 Interest rate measures

The paper uses two versions of the loan price:

- the **actual interest rate**, and
- the **standardized interest rate**, defined as the loan's actual rate divided by the cross-sectional standard deviation of rates in that calendar year.

The standardized measure is important because the People's Bank of China set bounds and base rates that changed over time. Raw rates remain informative, but year-to-year changes in dispersion make standardized rates the cleaner object for comparisons.

From Table I, mean actual rates are fairly similar across periods (7.00% pre-reform versus 6.89% post-reform), but the standard deviation of rates is larger after reform, which is why the standardized rate falls in mean terms from 12.99 to 7.47.

2.6 Ex post loan outcome

The main repayment outcome equals 1 if the loan is paid in full and on time at maturity and 0 otherwise. This is intentionally strict. A loan that is repaid only partially or with delay is counted as a bad outcome.

The authors also use a prior-default indicator among the borrower controls: this equals 1 if the borrower defaulted on a loan during the 12 months before the current loan application.

2.7 Communication-cost proxy

The post-reform data identify the loan officer responsible for the rating for 2,597 loans. For this subsample the paper observes:

- loan officer experience,
- branch president experience,
- **time worked together** for the officer-president pair,
- officer past performance for 2,186 loans.

The mean values from Table I are:

- loan officer experience: 3.47 years,
- branch president experience: 5.57 years,
- time worked together: 1.69 years,
- officer past performance: 0.88.

Interpretation. "Time worked together" is the paper's observable proxy for lower internal communication costs. The important point is that it is not merely officer experience or president experience; it is relationship-specific familiarity inside the branch hierarchy.

2.8 Outcome patterns before looking at regressions

Two descriptive facts matter.

First, Figure 1 shows that the rating distribution shifts after reform. Pre-reform, virtually nobody receives ratings 1 or 2; post-reform, low ratings appear and very high ratings become rarer. This is consistent with more conservative and more informative rating practices after personal accountability is introduced.

Second, repayment outcomes line up much better with ratings after reform. The paper notes that among firms with ratings 3 or better, pre-reform the probability of full and timely repayment rises only from 57.7% to 61.7% as ratings move from 3 to 8, whereas post-reform the gradient becomes much steeper and more monotone.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper tests two related hypotheses.

Hypothesis 1: incentive effect. After reform, internal ratings should become more tightly linked to loan pricing and to repayment because loan officers now bear greater responsibility for the ratings they produce.

Hypothesis 2: communication-cost effect. In the post-reform period, ratings should matter more when the officer and branch president have worked together longer, because information produced by the officer is easier for the final decision maker to understand and trust.

A third implication follows.

Hypothesis 3: overall information-use effect. If the bank as an organization becomes better informed, then the ex ante interest rate itself should become a better predictor of ex post loan outcomes after reform and when communication costs are lower.

3.2 Preliminary validation: ratings on hard information

Before running the main pricing and outcome regressions, the paper estimates reduced-form models of the internal rating on observable borrower characteristics:

$$\text{Rating}_{it} = \theta X_{it} + \text{fixed effects} + u_{it}. \tag{1}$$

This is not the structural model of interest. Its role is diagnostic. If ratings are higher quality after reform, they should line up better with hard information post-reform than pre-reform.

That is exactly what Table II shows: coefficients on log assets, leverage, ROA, and past default are much stronger and more sensible after reform, and the regression retains much more explanatory power even after stripping out city, year, and industry fixed effects.

3.3 Promotion regression as evidence on incentives

The paper also estimates a Probit model of whether a loan officer is promoted:

$$\Pr(\text{Promoted}_o = 1) = \Phi\big(\alpha + \beta\,\text{PastPerformance}_o + \gamma\,\text{Coastal}_o \times \text{PastPerformance}_o + Z_o'\delta\big)\,. \tag{2}$$

The point is not to explain promotions per se. The point is to show that officers with better ex post loan performance are more likely to advance, especially in coastal provinces. That makes the claim about incentive changes after reform more believable.

3.4 Main ex ante pricing regression

The first core specification asks whether the bank puts more weight on ratings after reform:

$$\text{InterestRate}_{it} = \beta_1\,\text{Rating}_{it} + \beta_2\,\text{Rating}_{it} \times \text{Post}_t + \text{FE} + X_{it}'\Gamma + (X_{it} \times \text{Post}_t)'\Lambda + \varepsilon_{it}. \tag{1}$$

The dependent variable is either the raw rate or the standardized rate.

Interpretation of signs.

- A higher rating means a safer borrower, so safer borrowers should pay lower rates.
- Therefore the effective pricing slope should be negative.
- If reform improves information quality, then $\beta_2 < 0$: the negative effect of rating on rates should become stronger after reform.

The coastal extension adds a triple interaction with a coastal-province indicator to test whether the reform had a larger effect where implementation and staff quality were stronger.

3.5 Communication-cost pricing regression

The second pricing specification is estimated only in the post-reform subsample where officer-president information is available:

$$\begin{aligned}
\text{InterestRate}_{it} = {} & \beta_1\,\text{Rating}_{it} + \beta_2\,\text{TimeWorkedTogether}_{it} && (3)\\
& + \beta_3\,\text{Rating}_{it} \times \text{TimeWorkedTogether}_{it} && (4)\\
& + \text{Experience terms and their interactions with rating} && (5)\\
& + \text{FE} + X_{it}'\Gamma + \varepsilon_{it}. && (2)
\end{aligned}$$

The experience controls include loan officer experience, branch president experience, and in some columns officer past performance.

Interpretation of signs.

- If lower communication costs make the bank use information more effectively, then $\beta_3 < 0$ in the interest-rate regression.
- The crucial comparison is between the pair-specific overlap variable and the generic experience variables. If only overlap matters, that points to communication rather than generic human capital.

3.6 Outcome regression using ratings

To test whether ratings actually forecast repayment better, the paper runs Probit models of the form:

$$\Pr(\text{Repay}_{it} = 1) = \Phi\left(\beta_1\,\text{Rating}_{it} + \beta_2\,\text{Rating}_{it} \times \text{Post}_t + \text{FE} + X_{it}'\Gamma + (X_{it} \times \text{Post}_t)'\Lambda\right). \quad (3a)$$

and, in the post-reform communication-cost sample,

$$\begin{aligned}
\Pr(\text{Repay}_{it} = 1) = \Phi\Big(& \beta_1\,\text{Rating}_{it} + \beta_2\,\text{TimeWorkedTogether}_{it} && (6)\\
& + \beta_3\,\text{Rating}_{it} \times \text{TimeWorkedTogether}_{it} + \text{experience terms} + \text{FE} + X_{it}'\Gamma\Big). && (3b)
\end{aligned}$$

Interpretation of signs.

- Safer ratings should predict a higher repayment probability, so the effective rating slope should be positive.
- If reform or lower communication costs improve information production, the marginal effect of rating on repayment should become more positive.

3.7 Outcome regression using interest rates

Finally, the paper replaces the rating with the loan interest rate in the repayment equation:

$$\Pr(\text{Repay}_{it} = 1) = \Phi\big(\alpha_1 \,\text{Rate}_{it} + \alpha_2 \,\text{Rate}_{it} \times \text{Post}_t + \text{FE} + X_{it}'\Pi + \cdots\big)\,, \tag{7}$$

and analogously with *Time worked together* in the post-reform subsample.

This is a strong diagnostic because the interest rate is supposed to summarize the bank's full assessment of risk. If rates predict repayment much better after reform, then it is not just the formal rating label that improved; the bank's overall information production and use improved.

3.8 Estimation choices

- Interest-rate regressions are estimated by OLS.
- Repayment and promotion regressions are estimated as Probit models, with marginal effects reported.
- Standard errors are clustered by borrower firm.
- All main regressions include year, city, and industry fixed effects.
- For specifications using reform timing, the direct effect of the post indicator is absorbed by year dummies, so the focus is on interaction terms.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main incentive estimate

The key source of identification is the timing of the organizational reform. The authors compare the same bank's lending behavior before and after decentralization, controlling for observable borrower quality and for city, year, and industry fixed effects.

Because the reform followed WTO entry and was imposed from above, the paper argues that it is plausibly exogenous from the perspective of the local loan officers producing ratings.

4.2 Why reverse causality is weak in the reform test

A single borrower or branch cannot plausibly cause the national reform. That rules out the most obvious reverse-causality concern. The more important concern is omitted time-varying factors: for example, the economy got stronger in the later period, borrowers were larger, and bank competition may have changed.

The paper addresses this by:

- including rich borrower controls,

- interacting borrower controls with the post-reform indicator,
- using city, year, and industry fixed effects,
- and later adding robustness controls for local bank competition and past lending relationships.

4.3 Why the communication-cost test is inherently weaker

The communication-cost measure, *Time worked together*, is not driven by an external policy shock. It is therefore not exogenous in the same clean way as the reform indicator.

The authors are very explicit about this limitation. They do not claim a perfect causal interpretation for the overlap variable. Instead, they try to rule out the most likely alternative explanations.

4.4 Endogenous assignment concern

A natural concern is that risky or opaque loans might be assigned to especially trusted officer-president pairs, mechanically making overlap look important.

Table V is the first check against this concern. Correlations between borrower observables and the experience/overlap variables are generally very small, almost always below 0.1 in absolute value. There is no strong pattern suggesting that better or worse borrowers are systematically assigned to pairs with longer overlap.

This is not a proof, but it is helpful evidence that assignment on observables is limited.

4.5 Why generic experience is not enough

If the overlap effect were really just a proxy for unobserved officer quality, then generic loan officer experience or branch president experience should matter in similar ways. But the results show that those generic experience variables do not explain the strengthened role of ratings nearly as well as *Time worked together* does.

That comparison is one of the most persuasive parts of the paper: what matters is not simply that the officer is seasoned or the president is seasoned; what matters is that the two have worked together.

4.6 Officer past performance as an additional check

In later specifications the authors add officer past performance and its interaction with the rating. This is meant to capture officer quality more directly. The interaction with past performance is not significant, and the main interaction with *Time worked together* remains very similar. Again, this does not solve the endogeneity problem fully, but it weakens the interpretation that overlap is merely disguising officer quality.

4.7 Other alternative explanations

The robustness section considers several competing stories.

Local banking competition. If more competition forces banks to produce better information, then the post-reform interaction could be proxying for market structure rather than internal reform. The paper adds the number of local lending institutions and interactions involving this variable; the main results remain.

Relationship lending. If the bank already knows the borrower well, then better pricing or prediction might reflect borrower-bank familiarity rather than internal reform or communication. The paper adds the number of past loans to the borrower and interactions with this relationship variable; again the main coefficients remain.

Ownership structure. Since Chinese SOEs may be treated differently, the paper re-estimates models after dropping SOEs and then within the SOE subsample. The core informational results remain, although some coastal and communication interactions become less sharp in the SOE-only sample.

Interest-rate bounds. Because regulated upper and lower bounds may compress rate variation, the paper also checks specifications excluding loans priced at the bounds. The results are similar.

4.8 What the paper can and cannot distinguish

The paper can distinguish reasonably well between:

- a world in which internal ratings are mostly uninformative and ignored,
- and a world in which internal ratings become informative and are used more heavily after incentive reform and when communication costs are lower.

The paper cannot fully distinguish between all possible micro-mechanisms inside the bank. For example, it cannot cleanly separate better truthful reporting from better processing of the same information, nor can it fully eliminate all nonrandom assignment concerns in the overlap test.

4.9 Relation to the literature

The paper sits at the intersection of several literatures:

- authority and incentives inside organizations,
- communication and hierarchy,
- information production in banking,
- and the role of distance and soft information in credit markets.

Its distinctive contribution is that it links a plausibly exogenous internal reform to both *pricing* and *outcomes*, while also bringing a direct proxy for organizational communication costs into the same empirical framework.

5 Tables and figures

5.1 Figure 1 and Table II: ratings become more conservative and more informative

Figure 1. The distribution of internal ratings shifts visibly after reform. Extremely high ratings become less common, and low ratings appear in the post-reform sample. The economic message is not that borrowers necessarily became worse; rather, officers appear less willing to inflate ratings once they can be held accountable.

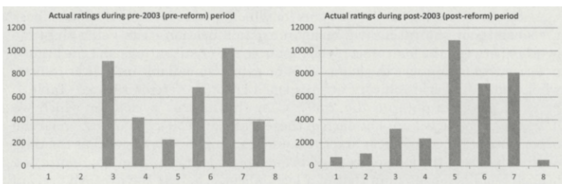

Figure 1. Distributions of actual internal ratings. In this figure, we plot histograms of actual internal ratings on loans during the pre-2003 (pre-reform) and post-2003 (post-reform) periods. There are 3,665 loans in the pre-2003 sample and 33,996 loans in the post-2003 sample. Internal ratings range from one to eight; higher ratings indicate higher credit quality.

Table II

Regressions of Credit Ratings on Hard Information Variables

We report OLS regression results of internal credit ratings on firm characteristics for both the pre-reform and the post-reform periods; the rating varies from one (riskiest) to eight (safest). The SOE dummy equals one when the borrower is a state-owned enterprise, and zero otherwise ("other ownership types" are the default type); the private enterprise dummy equals one when the borrower is a privately owned company, and zero otherwise. The previous default record indicator equals one when the borrower has defaulted on a loan during the 12 months prior to the application of the current loan, and zero otherwise. Standard errors are clustered by borrower firm. Robust *t*-statistics appear in parentheses below the coefficients; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

	Pre-Reform (1)	Post-Reform (2)	Pre-Reform (3)	Post-Reform (4)
Log assets	0.087**	0.403***	0.092**	0.356***
	(2.474)	(22.605)	(2.090)	(19.696)
Leverage	−1.293***	−1.574***	−1.412***	−1.849***
	(−5.401)	(−13.298)	(−5.124)	(−12.104)
ROA	1.728**	3.749***	1.764**	3.988***
	(2.162)	(12.815)	(2.058)	(11.924)
SOE	−0.202	−0.229***	−0.572***	−0.484***
	(−1.446)	(−3.635)	(−3.619)	(−6.700)
Private enterprise	−0.017	0.249***	−0.020	0.171***
	(−0.143)	(6.327)	(−0.143)	(3.686)
Previous default record	0.227*	−0.382***	0.403**	−0.457***
	(1.709)	(−8.900)	(2.563)	(−8.952)
Total asset turnover	0.080**	0.040	0.146***	0.023
	(2.171)	(1.428)	(3.966)	(0.979)
Year dummies	Yes	Yes	No	No
City dummies	Yes	Yes	No	No
Industry dummies	Yes	Yes	No	No
Observations	3,665	33,996	3,665	33,996
Adjusted-R^2	0.378	0.389	0.083	0.229

Table II. Regressing ratings on hard information shows that post-reform ratings line up much more strongly with borrower fundamentals. For example:

- log assets: 0.087 pre-reform versus 0.403 post-reform,
- ROA: 1.728 versus 3.749,
- previous default: positive and weak pre-reform, strongly negative post-reform.

The biggest conceptual message is that ratings become tied to actual borrower condition after reform rather than mostly reflecting fixed effects and committee noise.

5.2 Table III: incentives are real because promotions reward performance

Table III validates the incentive story. Officers whose past loan portfolios perform better are more likely to be promoted, and the link is stronger in coastal provinces. A one-standard-deviation increase in past performance raises promotion probability by about 23 percentage points, and by about 32 percentage points in coastal areas.

This is not one of the main outcome tables, but it is important because it shows that the post-reform organization created meaningful career incentives.

Table III

Regressions of the Likelihood of Promotion on Loan Officers' Past Performance

This table reports marginal effects from Probit regressions of *Promoted*, which equals one if a loan officer was promoted to a higher position (similar ranking as a branch president) within the same branch or moves from a lower-level branch to a higher-level one (holding a similar or higher position), and zero otherwise. Average past performance is the percentage of loans fully repaid on time out of all (post-reform) loans processed by the loan officer before the promotion; for those *not* promoted, we use the average performance of all loans made by the officer through the end of the sample. Average credit rating equals the average rating on borrower firms processed by the loan officer in the post-reform period; coastal equals one for branches located in coastal regions, and zero for inland branches; length of service equals the log of the number of days the officer had been working at the time of the promotion. Data are from the post-reform period only (for 299 officers), because we can only observe loan officer performance after the reform. Robust z-statistics are in parentheses below the coefficients; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

	(1)	(2)	(3)	(4)
Average past performance	1.082***	1.128***	0.762***	−0.095
	(5.268)	(5.363)	(2.917)	(−0.134)
Coastal*Average past performance	–	–	0.822**	0.916**
	–	–	(2.036)	(2.173)
Average credit rating	−0.122***	−0.123***	−0.124***	−0.278**
	(−4.258)	(−4.284)	(−4.371)	(−2.260)
Average past performance*	–	–	–	0.181
Average credit rating	–	–	–	(1.308)
Length of service	–	0.048	0.053	0.054
	–	(1.349)	(1.493)	(1.502)
City dummies	Yes	Yes	Yes	Yes
Observations	299	299	299	299
Pseudo-R^2	0.316	0.321	0.332	0.337
Past performance				
Mean		0.88		
SD		0.20		

Table IV

Regression of Ex Ante Loan Terms on Credit Rating: Pre-Reform versus Post-Reform

We report OLS regression results of loan interest rates (actual and standardized rates) on the internal credit rating and its interaction with the post-reform dummy; the rating varies from one (riskiest) to eight (safest). The standardized interest rate is the actual interest rate on a loan over the standard deviation of rates on all loans in a single year. The post-reform dummy equals one when the loan is made after the reform, defined as in Table I; coastal equals one for branches located in coastal regions, and zero for inland branches. Branch size is total deposits (in billions of RMB) in the initial year of our sample period (2000). Firm controls and interactions include financial variables (firm size, leverage, ROA), ownership types (SOE), previous default record, total asset turnover ratio, and interactions between the post-reform indicator and the financial variables. Standard errors are clustered by borrower firm. Robust t-statistics appear in parentheses below the coefficients; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

	Actual Interest Rate (1)	Standardized Interest Rate (2)	Actual Interest Rate (3)	Standardized Interest Rate (4)
Credit rating	0.010	0.009	−0.022	−0.006
	(0.694)	(0.504)	(−1.534)	(−0.351)
Post*Credit rating	−0.080***	−0.087***	−0.037***	−0.063***
	(−5.824)	(−4.920)	(−2.633)	(−3.427)
Coastal*Credit rating	–	–	0.035***	−0.003
	–	–	(2.612)	(−0.154)
Coastal*Post*Credit rating	–	–	−0.082***	−0.036***
	–	–	(−7.887)	(−2.671)
Branch size	−0.069	−0.109	−0.163**	−0.203**
	(−0.962)	(−1.333)	(−2.161)	(−2.382)
Branch size*Credit rating	0.002	0.010	0.020	0.027*
	(0.180)	(0.619)	(1.416)	(1.695)
Log assets	−0.195***	−0.220***	−0.196***	−0.220***
	(−20.891)	(−21.432)	(−21.017)	(−21.453)
Leverage	−0.127**	−0.146**	−0.128**	−0.145**
	(−2.415)	(−2.483)	(−2.428)	(−2.465)
ROA	0.122	0.023	0.136	0.029
	(0.866)	(0.157)	(0.975)	(0.195)
SOE	0.059	−0.122*	0.103**	−0.106
	(1.141)	(−1.696)	(1.988)	(−1.470)
Private enterprise	0.155***	0.167***	0.163***	0.171***
	(7.404)	(7.491)	(7.780)	(7.720)
Previous default record	0.147***	0.152***	0.135***	0.147***
	(7.376)	(7.066)	(6.882)	(6.923)
Total asset turnover	−0.031***	−0.031***	−0.030***	−0.031***
	(−3.356)	(−3.232)	(−3.245)	(−3.151)
Post*Firm controls	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes
City dummies	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes
Observations	37,661	37,661	37,661	37,661
Adjusted-R^2	0.545	0.887	0.548	0.887

5.3 Table IV: reform makes ratings matter for pricing

Table IV is the central reform table for loan pricing. In the standardized-rate specification with all controls and coastal interactions, the key coefficients are:

- **Post** $\times$ **Credit rating** $= -0.063^{***}$,
- **Coastal** $\times$ **Post** $\times$ **Credit rating** $= -0.036^{***}$.

Reading: before reform, the rating by itself has little marginal pricing power once controls and fixed effects are included; after reform, higher ratings lead to materially lower loan rates, and this incremental effect is even stronger in coastal provinces.

Economic message: the bank begins using its internal signal much more aggressively in loan pricing once responsibility is individualized.

5.4 Table V and Table VI: communication costs matter above and beyond experience

Table V. Borrower characteristics are only weakly correlated with loan officer experience, branch president experience, and the officer-president overlap measure. This supports the claim that the communication-cost proxy is not obviously just capturing selection on observables.

Table VI. This is the key communication-cost pricing table. The most informative coefficients are in the standardized-rate regressions:

Table V

Correlations between Work Experience and Borrower Characteristics

This table reports simple correlations between loan officer experience, branch president experience, the time the two have worked together in the same branch, and loan officer past performance with borrower characteristics and credit ratings. The sample is 2,597 loans from the post-reform period (2,186 for the sample with loan officer past performance).

	Log Asset	Leverage	ROA	SOE	Private	Default Record	Total Asset Turnover	Credit Rating
Loan officer experience	−0.057	0.011	0.030	−0.005	−0.018	−0.045	−0.041	0.045
Branch president experience	−0.250	−0.118	0.201	−0.113	0.109	−0.156	0.089	−0.098
Time worked together	−0.078	−0.043	0.040	−0.002	−0.015	−0.109	−0.047	0.083
Loan officer past performance	−0.087	−0.054	0.071	−0.074	0.067	−0.081	0.068	0.019

Table VI

Regression of Ex Ante Loan Terms on Credit Rating: The Effect of Communication Costs in the Post-Reform Period

We report OLS regression results of loan interest rates (actual and standardized) on credit rating and interactions with loan officer experience (in years), branch president experience (in years), and time worked together of the loan officer and president (in years). The sample includes the post-reform period only; ratings vary from one (riskiest) to eight (safest). Branch size is total deposits (in billions of RMB) in the initial year of our sample period (2000) for each branch. The standardized interest rate is the interest rate on a loan over the standard deviation of rates on all loans in a given year. Officer past performance is measured at the loan level: for each loan, we compute the fraction of loans made by the same officer prior to loan origination that subsequently default. Standard errors are clustered by borrower firm. Robust t-statistics appear in parentheses below the coefficients; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Actual Rate (1)	Std. Rate (2)	Actual Rate (3)	Std. Rate (4)	Actual Rate (5)	Std. Rate (6)	Actual Rate (7)	Std. Rate (8)
Credit rating	−0.091***	−0.088***	−0.025	−0.028	0.000	−0.000	−0.027	−0.011
	(−4.645)	(−4.626)	(−0.500)	(−0.567)	(0.006)	(−0.008)	(−0.329)	(−0.134)
Time worked together*	–	–	−0.034***	−0.033***	−0.030**	−0.028**	−0.030**	−0.029**
Credit rating	–	–	(−2.688)	(−2.637)	(−2.088)	(−2.042)	(−2.106)	(−2.091)
Loan officer experience*	–	–	–	–	−0.001	−0.001	−0.002	−0.000
Credit rating	–	–	–	–	(−0.076)	(−0.055)	(−0.149)	(−0.045)
Branch president experience*	–	–	–	–	−0.007	−0.007	−0.007	−0.007
Credit rating	–	–	–	–	(−1.400)	(−1.595)	(−1.479)	(−1.625)
Officer past performance	–	–	–	–	–	–	−0.219	−0.131
	–	–	–	–	–	–	(−0.548)	(−0.344)
Officer past performance*	–	–	–	–	–	–	0.039	0.020
Credit rating	–	–	–	–	–	–	(0.519)	(0.282)
Time worked together	−0.085***	−0.079***	0.090	0.088	0.067	0.064	0.070	0.068
	(−3.026)	(−2.901)	(1.319)	(1.328)	(0.859)	(0.856)	(0.892)	(0.906)
Loan officer experience	−0.023	−0.022	−0.020	−0.019	−0.015	−0.015	−0.010	−0.015
	(−1.303)	(−1.289)	(−1.137)	(−1.121)	(−0.259)	(−0.269)	(−0.180)	(−0.272)

(Continued)

Table VI—*Continued*

	Actual Rate (1)	Std. Rate (2)	Actual Rate (3)	Std. Rate (4)	Actual Rate (5)	Std. Rate (6)	Actual Rate (7)	Std. Rate (8)
Branch president experience	−0.007	−0.007	−0.007	−0.007	0.028	0.032	0.030	0.033
	(−0.756)	(−0.775)	(−0.696)	(−0.710)	(1.121)	(1.314)	(1.195)	(1.350)
Branch size	−0.109	−0.095	−0.079	−0.088	−0.138	−0.154	−0.156	0.104
	(−1.055)	(−0.941)	(−0.330)	(−0.377)	(−0.547)	(−0.625)	(−0.618)	(0.276)
Branch size*	–	–	−0.010	−0.006	−0.001	0.004	0.003	0.007
Credit rating	–	–	(−0.228)	(−0.136)	(−0.020)	(0.096)	(0.053)	(0.155)
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,597	2,597	2,597	2,597	2,597	2,597	2,597	2,597
Adjusted-R^2	0.503	0.625	0.507	0.628	0.509	0.630	0.508	0.503

- column 2: **Credit rating** $= -0.088^{***}$,
- column 4: **Time worked together** $\times$ **Credit rating** $= -0.033^{***}$,
- column 8: with officer past performance added, the interaction remains -0.029^{**}.

The paper's own calculation is especially useful. At the mean overlap of 1.69 years, a four-notch increase in rating lowers the standardized rate by about 0.34. If overlap is one standard deviation higher, the same four-notch rating improvement lowers the standardized rate by about 0.52.

Economic message: not only does the bank use ratings more after reform; it uses them especially strongly when the information producer and decision maker are organizationally closer.

5.5 Table VII and Table VIII: ratings forecast outcomes much better after reform and with lower communication costs

Table VII

Summary Statistics on Ex Post Loan Performance

This table reports simple default statistics sorted by internal credit ratings, divided into pre- and post-reform regimes. Pay off on time means the borrower pays off the entire loan on or before the maturity date, otherwise the borrower is in default, using one year after the original loan maturity date as the cutoff date for (late) repayment. Credit ratings vary from one (riskiest) to eight (safest).

		Pre-Reform (Before 2003)								Post-Reform (After 2003)							
Credit Rating		1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8
Breach of contract (%)	Did not repay	0	–	0.6	0.7	0	0.6	0.4	2.3	1.1	1.4	1.3	2.3	1.7	1.9	1.3	0
	Partial repay	75	–	31.9	37.7	42.8	46.4	38.3	31.1	24.7	29	24	16.4	11.4	5.7	3.3	0
	Pay off late	25	–	9.9	8.8	7.9	6.3	5.3	4.9	2.1	2.5	1.9	1.1	0.7	0.3	0.3	0
Contract performance (%)	Pay off on time	0	–	57.7	52.8	49.3	46.7	56.1	61.7	72.1	67	72.8	80.3	86.1	92.2	95.2	100
Number of loans		4	0	912	422	229	685	1,024	389	753	1,071	3,201	2,364	10,845	7,160	8,077	525

Table VIII

Regression of Ex Post Loan Outcomes on Credit Rating

In columns 1 and 2, we report marginal effects from Probit regressions of loan performance on the credit rating and its interactions with the post-reform indicator and a coastal indicator; ratings vary from one (riskiest) to eight (safest). In columns 3 and 4, we report marginal effects from Probit regressions of loan performance on the credit rating and its interaction with loan officer experience (in years), branch president experience (in years), and the time working together of the loan officer and branch president (in years); officer past performance (measured at the loan level) is the fraction of loans made by the same officer prior to loan origination that subsequently default. Branch size is total deposits (in billions of RMB) in the initial year (2000) of our sample period. Firm controls include financial variables (firm size, leverage, ROA), ownership type (SOE), previous default record, and total asset turnover ratio. The dependent variable equals one if a loan is paid in full at the maturity date and zero otherwise. Standard errors are clustered by borrower firm. Robust z-statistics appear in parentheses below the coefficients; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Loan Outcome (Pre-Reform vs. Post-Reform)		Loan Outcome (Post-Reform)	
	(1)	(2)	(3)	(4)
Credit rating	−0.000	0.003	0.014	0.029
	(−0.082)	(0.617)	(0.966)	(0.958)
Post*Credit rating	0.025***	0.019***	–	–
	(5.357)	(3.928)	–	–
Coastal*	–	0.011**	–	–
Credit rating	–	(2.119)	–	–
Coastal*Post*	–	0.006	–	–
Credit rating	–	(1.621)	–	–
Time worked together*	–	–	0.009**	0.008**
Credit rating	–	–	(2.194)	(2.082)
Loan officer experience*	–	–	0.000	0.001
Credit rating	–	–	(0.005)	(0.288)
Branch president experience*	–	–	−0.002	−0.002*
Credit rating	–	–	(−1.594)	(−1.700)
Time worked together	–	–	−0.023	−0.019
	–	–	(−1.143)	(−0.994)
Loan officer experience	–	–	−0.006	−0.010
	–	–	(−0.391)	(−0.653)
Branch president experience	–	–	0.001	0.001
	–	–	(0.169)	(0.226)
Officer past performance	–	–	–	0.176
	–	–	–	(1.189)
Officer past performance*	–	–	–	−0.019
Credit rating	–	–	–	(−0.642)
Branch size	−0.023	0.019	−0.036	−0.022
	(−1.002)	(0.741)	(−0.600)	(−0.363)
Branch size*	0.011***	0.003	0.012	0.010
Credit rating	(2.637)	(0.569)	(0.974)	(0.786)
Firm controls	Yes	Yes	Yes	Yes
Post*Firm controls	Yes	Yes	–	–

(Continued)

Table VIII—*Continued*

	Loan Outcome (Pre-Reform vs. Post-Reform)		Loan Outcome (Post-Reform)	
	(1)	(2)	(3)	(4)
Year dummies	Yes	Yes	Yes	Yes
City dummies	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes
Observations	37,661	37,661	2,416	2,416
Pseudo-R^2	0.190	0.191	0.193	0.199

Table VII. Simple descriptive default rates show that the repayment gradient by rating becomes much steeper after reform. Pre-reform, repayment differences across ratings are weak and sometimes nonmonotonic. Post-reform, better ratings are much more clearly associated with full and timely repayment.

Table VIII. In the formal Probit outcome regressions:

- **Post** $\times$ **Credit rating** $= 0.025^{***}$ in column 1 and 0.019^{***} in column 2,
- **Time worked together** $\times$ **Credit rating** $= 0.009^{**}$ in column 3 and 0.008^{**} in column 4.

Reading: a one-notch improvement in the rating raises the probability of full repayment by roughly 2.5 percentage points after reform. In the post-reform subsample, ratings forecast repayment much better when overlap is longer.

Economic message: this table validates the pricing results using actual repayment outcomes. The bank is not merely pretending to use better information; the information is genuinely more predictive.

5.6 Table IX: the bank's overall information, not just the formal rating, improves

Table IX

Regression of Ex Post Loan Outcome on Ex Ante Interest Rate

We report marginal effects from Probit regressions of loan outcomes on the standardized interest rate (interest rate/sample standard deviation of rates for that year) and interactions with the post-reform and coastal indicators (columns 1 and 2) and time worked together (columns 3 and 4). The dependent variable equals one if a loan is paid in full at the maturity date and zero otherwise. The post-reform dummy equals one when the loan is made after the reform (post-2003) and zero otherwise; officer past performance (measured at the loan level) is the fraction of loans made by the same officer prior to loan origination that subsequently default. We include the following firm control variables: financial variables (firm size, leverage, ROA), ownership types (SOE, private), previous default record, total asset turnover ratio, and the interactions between the post-reform indicator and the financial variables. Branch size is total deposits (billions of RMB) in the initial year (2000) of our sample period for each branch. Standard errors are clustered by borrower firm. Robust z-statistics appear in parentheses below the coefficients; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

	Loan Outcome (Pre-Reform vs. Post-Reform)		Loan Outcome (Post-Reform)	
	(1)	(2)	(3)	(4)
Std. interest rate	−0.001	0.004	−0.022	0.022
	(−0.107)	(0.486)	(−0.880)	(0.411)
Post*Std. interest rate	−0.026***	−0.030***	–	–
	(−3.301)	(−3.683)	–	–
Coastal*Std. interest rate	–	0.006	–	–
	–	(1.520)	–	–
Coastal*Post*	–	−0.005**	–	–
Std. interest rate	–	(−2.097)	–	–
Time worked together*	–	–	−0.016**	−0.014**
Std. interest rate	–	–	(−2.157)	(−2.028)
Loan officer experience*	–	–	−0.003	−0.002
Std. interest rate	–	–	(−0.473)	(−0.422)
Branch president experience*	–	–	0.003*	0.002
Std. interest rate	–	–	(1.699)	(1.569)
Time worked together	–	–	0.118**	0.110**
	–	–	(2.491)	(2.363)
Loan officer experience	–	–	0.011	0.009
	–	–	(0.319)	(0.265)
Branch president experience	–	–	−0.024**	−0.023**
	–	–	(−2.300)	(−2.182)
Officer past performance	–	–	–	0.408
	–	–	–	(1.234)
Officer past performance*	–	–	–	−0.049
Credit rating	–	–	–	(−0.992)
Branch size	0.112***	0.142***	0.201	0.207
	(3.905)	(4.469)	(1.578)	(1.603)
Branch size*	−0.010***	−0.014***	−0.027	−0.028
Std. interest rate	(−3.090)	(−3.741)	(−1.418)	(−1.416)
Firm controls	Yes	Yes	Yes	Yes
Post*Firm controls	Yes	Yes	–	–

(Continued)

Table IX—*Continued*

	Loan Outcome (Pre-Reform vs. Post-Reform)		Loan Outcome (Post-Reform)	
	(1)	(2)	(3)	(4)
Year dummies	Yes	Yes	Yes	Yes
City dummies	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes
Observations	37,661	37,661	2,416	2,416
Pseudo-R^2	0.178	0.179	0.205	0.211

Table IX asks whether the loan interest rate itself becomes more predictive of repayment. This is powerful because the rate should summarize the bank's overall risk assessment.

The main coefficients are:

- **Post** $\times$ **Standardized interest rate** $= -0.026^{***}$ in column 1 and -0.030^{***} in column 2,
- **Coastal** $\times$ **Post** $\times$ **Standardized interest rate** $= -0.005^{**}$,
- **Time worked together** $\times$ **Standardized interest rate** $= -0.016^{**}$ in column 3 and -0.014^{**} in column 4.

Reading: after reform, higher ex ante interest rates are much more strongly associated with default. The same is true when overlap is longer.

Economic message: the whole internal information system improves. It is not just the documented rating that becomes better; the final loan contract embeds better risk information.

5.7 Table X and the appendix case studies: the story survives the obvious objections

Table X

Robustness Tests

This table reports robustness tests for links between the credit rating and standardized interest rates (the interest rate standardized by the cross-sectional standard deviation in that year) and the loan outcome (equals one if full repayment and zero otherwise). Credit ratings vary from one (riskiest) to eight (safest). We only report the key interaction term from the earlier models, so each coefficient below represents one regression. Panel A reproduces the coefficients of interest from our baseline models (Table IV, column 4; Table V, column 8; Table VIII, columns 2 and 4; Table IX, columns 2 and 4). Panel B adds the number of lending institutions in the same local market as the branch making the loan and its interaction with the post-2003 indicator or time worked together. Panel C adds the number of past loans made with the borrower and its interaction with the post-2003 indicator or time worked together. Panel D drops SOEs from the sample. Panel E reports results for the sample of SOEs. Standard errors are clustered by borrower. Robust z-statistics or t-statistics appear in parentheses below the coefficients; ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

		Std. Interest Rate	Loan Outcome	
		(1)	(2)	
Panel A. Prior Results				
Post*Credit rating		−0.063***	0.019***	
	Table IV, col. 4 →	(−3.427)	(3.928)	← Table VIII, col. 2
Coastal*Post*Credit rating		−0.036***	0.006	
		(−2.671)	(1.621)	
Time worked together*Credit rating	Table VI, col. 8 →	−0.029**	0.008**	
		(−2.091)	(2.082)	← Table VIII, col. 4
Post*Std. interest rate		–	−0.030***	
		–	(−3.683)	
Coastal*Post*Std. interest rate		–	−0.005**	← Table IX, col. 2
		–	(−2.097)	
Time worked together*Std. interest rate		–	−0.014**	
		–	(−2.028)	← Table IX, col. 4
Panel B. Control for Number of Competing Banks				
Post*Credit rating		−0.063***	0.019***	
		(−3.473)	(3.884)	
Coastal*Post*Credit rating		−0.034**	0.006	
		(−2.539)	(1.610)	
Time worked together*Credit rating		−0.028**	0.008**	
		(−2.024)	(2.007)	
Post*Std. interest rate		–	−0.033***	
		–	(−3.935)	
Coastal*Post*Std. interest rate		–	−0.005*	
		–	(−1.935)	
Time worked together*Std. interest rate		–	−0.013*	
		–	(−1.758)	
Panel C. Control for the Number of Past Loans				
Post*Credit rating		−0.032*	0.019***	
		(−1.783)	(3.834)	
Coastal*Post*Credit rating		−0.040***	0.005	
		(−2.976)	(1.269)	
Time worked together*Credit rating		−0.030**	0.006*	
		(−2.087)	(1.700)	
Post*Std. interest rate		–	−0.030***	
		–	(−3.666)	
Coastal*Post*Std. interest rate		–	−0.004*	
		–	(−1.660)	
Time worked together*Std. interest rate		–	−0.013*	
		–	(−1.862)	
Panel D . Drop SOEs				
Post*Credit rating		−0.046**	0.013**	
		(−2.214)	(2.533)	
Coastal*Post*Credit rating		−0.049***	0.006	
		(−3.363)	(1.604)	
Time worked together*Credit rating		−0.032*	0.005	
		(−1.885)	(1.298)	
Post*Std. interest rate		–	−0.028***	
		–	(−2.970)	
Coastal*Post*Std. interest rate		–	−0.007***	
		–	(−2.739)	
Time worked together*Std. interest rate		–	−0.016**	
		–	(−2.147)	
Panel E. SOEs				
Post*Credit rating		−.092***	0.041***	
		(−2.590)	(3.724)	
Coastal*Post*Credit rating		0.028	0.004	
		(0.661)	(0.436)	
Time worked together*Credit rating		0.002	0.019	
		(0.075)	(1.518)	
Post*Std. interest rate		–	−0.045***	
		–	(−2.637)	
Coastal*Post*Std. interest rate		–	−0.001	
		–	(−0.178)	
Time worked together*Std. interest rate		–	0.043**	
		–	(1.960)	

(*Continued*)

Table X—*Continued*

Table X shows that the core interactions survive controls for local lending competition, past borrower-bank relationships, and alternative ownership samples. The post-reform and communication-cost effects remain economically similar and usually statistically significant.

The appendix case studies help interpret what is going on. One case illustrates productive use of soft information, where officers learn about a credible restructuring plan that is not publicly verifiable and the firm eventually repays. The other illustrates how political pressure can distort rating production and lead to a bad loan. These cases nicely reinforce the paper's central message: soft information is valuable, but only if the organization's incentives and internal communication process allow it to be produced and used well.

6 Short summary

The paper shows that internal bank organization shapes credit-market outcomes. A reform that delegated responsibility to individual loan officers after China's WTO entry made internal ratings more informative and made the bank rely more on those ratings in loan pricing. In the post-reform sample, the same pattern appears cross-sectionally: ratings matter more when the officer producing the information and the branch president approving the loan have worked together longer. Ratings and interest rates also become better predictors of repayment in exactly those situations. The evidence is strongest for the reform-based comparisons and suggestive but still persuasive for the communication-cost channel. Overall, the paper provides unusually direct evidence that incentives and communication costs affect information production, transmission, and use inside financial institutions.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the quality and use of bank-generated information are not fixed technological objects. They depend on who is responsible for producing that information and on how easily it can be transmitted to the person making the final decision. When responsibility becomes individualized and when internal communication costs are lower, credit ratings receive more weight in pricing and predict loan outcomes better.

7.2 Economic intuition

Soft information is valuable but fragile. It is costly to gather, easy to distort, and hard to transmit through a hierarchy. The reform improves the incentive to produce truthful and careful information. Longer officer-president overlap lowers the loss that occurs when that information has to move inside the organization. Better information then appears in both rates and default outcomes.

7.3 Takeaway

This paper demonstrates that organizational design matters for finance. Credit allocation is not determined only by borrower fundamentals and bank balance sheets; it also depends on the internal incentive system and communication network of the lender.